Charpeter 1

MLE

Q1:

→ Given

- Sequence:
  - $y_2 = ay_1 + \delta x + w_2$
  - $y_1 = ay_0 + \delta x + w_1$
  - $y_0 = \delta x + w_0$
- w_0, w_1 and w_2 are i.i.d. with the distribution $N(0, \sigma^2)$
- a and δ are unknown.
- $p(y_2, y_1, y_0 \mid y_1, y_0, x; a)$

→ Find

- Compute for value of a and δ that maximizes $p(y_0, y_1, y_2 \mid y_1, y_0, x; a)$

Solution

Step 1 : isolate noise

from $y_2 = ay_1 + \delta x + w_2$ to $w_2 = y_2 - ay_1 - \delta x$

from $y_1 = ay_0 + \delta x + w_1$ to $w_1 = y_1 - ay_0 - \delta x$

from $y_0 = \delta x + w_0$ to $w_0 = y_0 - \delta x$

∴ In this step we get

$$w_2 = y_2 - ay_1 - \delta x$$
$$w_1 = y_1 - ay_0 - \delta x$$
$$w_0 = y_0 - \delta x$$

Step 2 : Formulate PDF of $N(0, \sigma^2)$

since we know PDF of $N(\mu, \sigma^2)$ is

$$f(w_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\left(-\frac{(w_i - \mu)^2}{2\sigma^2}\right)}$$

since we know $\mu = 0$ we get

$$f(w_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\left(-\frac{w_i^2}{2\sigma^2}\right)}$$

∴ In this step we get PDF of $N(0, \sigma^2)$

$$f(w_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\left(-\frac{w_i^2}{2\sigma^2}\right)}$$

Step 3 : Formulate Likelihood function (lik)

$$\text{lik}(a, \delta) = \prod_{i=1}^{2} f(w_i)$$
$$= \prod_{i=1}^{2} \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\left(-\frac{w_i^2}{2\sigma^2}\right)}$$

step 4: Formulate log-Likelihood function ($\mathcal{L}$) using natural-log

$$\mathcal{L}(a,\delta) = \ln(\text{lik}(a,\delta)) = \ln \prod_{i=1}^{2} \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\left(-\frac{w_i^2}{2\sigma^2}\right)}$$

$$= \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) + \ln\left(\prod_{i=1}^{2} e^{\left(-\frac{w_i^2}{2\sigma^2}\right)}\right)$$

$$= \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) + \sum_{i=1}^{2} \ln\left(e^{\left(-\frac{w_i^2}{2\sigma^2}\right)}\right)$$

$$= \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2}\sum_{i=1}^{2} w_i^2$$

$$\rightarrow \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2}\left[w_0^2 + w_1^2 + w_2^2\right]$$

step 5: Maximizing $\underset{a,\delta}{\text{argmax}}(\mathcal{L}(a,\delta))$

Let look at $\mathcal{L}(a,\delta) = \underbrace{\ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)}_{\text{constant}} - \underbrace{\frac{1}{2\sigma^2}}_{\text{constant}}\underbrace{\left[w_0^2 + w_1^2 + w_2^2\right]}_{\text{variable}}$

we can see that variable part is multipy by -1
to maximize this we have to minimize a variable part

$$\rightarrow \frac{\partial \mathcal{L}(a,\delta)}{\partial a} = 0 \quad \text{and} \quad \frac{\partial \mathcal{L}(a,\delta)}{\partial \delta} = 0$$

we substitute w_0, w_1, w_2 we get

$$\frac{\partial \mathcal{L}(a,\delta)}{\partial a} = \frac{\partial\left[(y_0 - \delta x)^2 + (y_1 - ay_0 - \delta x)^2 + (y_2 - ay_1 - \delta x)^2\right]}{\partial a} = 0$$

$$y_0 y_1 + y_1 y_2 = a(y_0^2 + y_1^2) + \delta x(y_0 + y_1) \longrightarrow (1)$$

$$\frac{\partial \mathcal{L}(a,\delta)}{\partial \delta} = \frac{\partial\left[(y_0 - \delta x)^2 + (y_1 - ay_0 - \delta x)^2 + (y_2 - ay_1 - \delta x)^2\right]}{\partial \delta} = 0$$

$$y_0 + y_1 + y_2 = a(y_0 + y_1) + 3\delta x \longrightarrow (2)$$

$$\hat{\delta} = \frac{(y_0 + y_1 + y_2) - a(y_0 + y_1)}{3x} \longrightarrow (3)$$

solve for a by subtitute (3) into (1)

$$\hat{a} = \frac{y_0 y_1 + 2y_1 y_2 - y_0^2 - y_1^2 - y_0 y_2}{2y_0^2 + 2y_1^2 - 2y_0 y_1}$$

Simple Bayes Classifier

Q2:

→ Given:
- Class
  - 1 : Happy Cat
  - 2 : Sad Cat
- Posterior probabilities
  - $P(x|w_1) = N(5,2)$
  - $P(x|w_2) = N(0,2)$
- $P(w_1) = P(w_2) = 0.5$

→ Find:
- Plot the posterior probabilites of to class on the same axis.
- Determin the decision boundary.
- What happen with $x < 2$. Is it Happy Cat or Sad cat

2.1

<u>step 1</u>: Formulate PDF of $N(5,2)$ and $N(0,2)$

As we know PDF of $N(\mu, \sigma^2)$ is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

we also know μ, σ^2 of Happy cat and Sad cat

Happy Cat : $(\mu, \sigma^2) = (5,2)$
Sad Cat : $(\mu, \sigma^2) = (0,2)$

we can formulate PDF of Happy cat and Sad cat

PDF of Happy cat : $\frac{1}{\sqrt{4\pi}} \cdot e^{\left(-\frac{(x-5)^2}{4}\right)}$

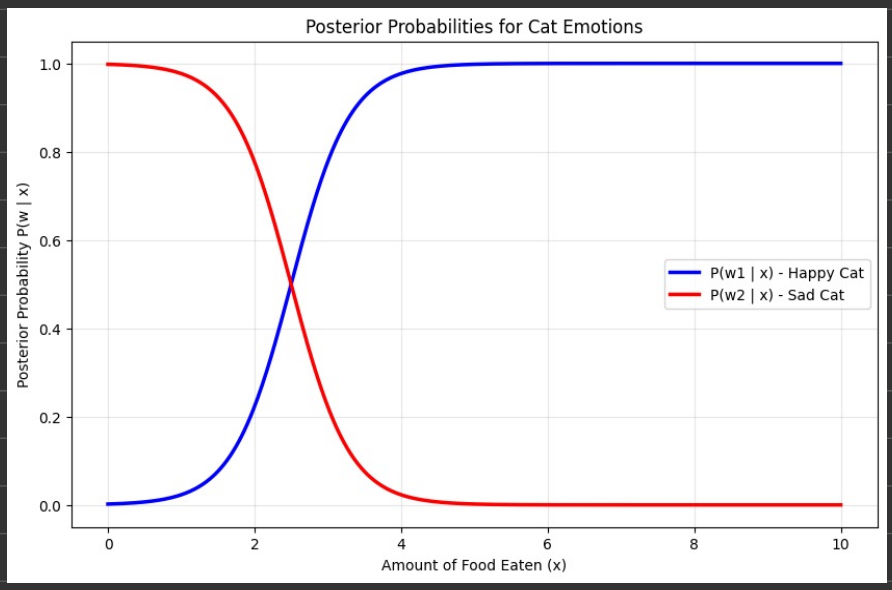
PDF of Sad cat : $\frac{1}{\sqrt{4\pi}} \cdot e^{\left(-\frac{x^2}{4}\right)}$

<u>step 2</u>: Calculat the posterior probability function

From Bayes Theory : $P(w_i|x) = \frac{P(x|w_i)P(w_i)}{P(x)}$

and $p(x) = p(x|w_1)p(w_1) + p(x|w_2)p(w_2)$

step 3: Plot the posterior probability by using $x = [0, 10]$



2.2

step 1: Formulate likelihood ratio test

$$\frac{P(x \mid \omega_1)}{P(x \mid \omega_2)} = \frac{P(\omega_2)}{P(\omega_1)}$$

substitus Gaussian likelihood, we get

$$\frac{\frac{1}{\sqrt{4\pi}} \cdot e^{\left(-\frac{(x-5)^2}{4}\right)}}{\frac{1}{\sqrt{4\pi}} \cdot e^{\left(-\frac{x^2}{4}\right)}} = \frac{0.5}{0.5} = 1$$

$$\rightarrow \cancel{\frac{1}{\sqrt{4\pi}}} \cdot e^{\left(-\frac{(x-5)^2}{4}\right)} = \cancel{\frac{1}{\sqrt{4\pi}}} \cdot e^{\left(-\frac{x^2}{4}\right)}$$

$$\rightarrow e^{\left(-\frac{(x-5)^2}{4}\right)} = e^{\left(-\frac{x^2}{4}\right)}$$

step 2: Take ln in both side

$$\ln\left(e^{\left(-\frac{(x-5)^2}{4}\right)}\right) = \left(e^{\left(-\frac{x^2}{4}\right)}\right)$$

$$\rightarrow x^2 - 10x + 25 = x^2$$

step 3: Solve for x

$$x^2 - 10x + 25 = x^2$$

$$10x = 25$$

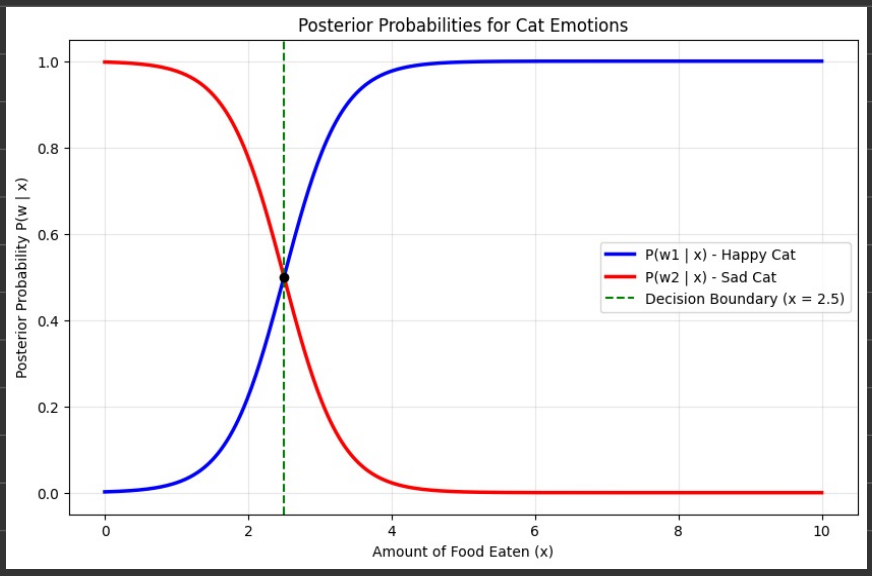
$$x = 2.5$$

$\therefore$ The decision boundary is at $x = 2.5$

2.3

What if $x < 2$ what will happend is it going to be sad cat or Happy cat

Step 1: Plot the posterior probability with decision boundary



Step 2: Define Region

As we can see there are 3 Regions

Region 1 ($x < 2.5$) $P(w_2) > P(w_1)$ → Sad Cat
Region 2 ($x > 2.5$) $P(w_1) > P(w_2)$ → Happy cat
Region 3 ($x = 2.5$) $P(w_1) = P(w_2)$ Decise by your self

If it $x < 2$ It mean it at Region 1 ($x < 2.5$)
It is definitely a Sad cat

Q3

→ Given : • $P(\omega_1) = 0.8$ (Happy Cat)
• $P(\omega_2) = 0.2$ (Sad Cat)

→ Find : • Determind Decision boundary
• What will happend when $x < 2$

step1: Formulate and subtitute a likelihood ratio test

$$\frac{P(x|\omega_1)}{P(x|\omega_2)} = \frac{P(\omega_2)}{P(\omega_1)}$$

$$\rightarrow \frac{\cancel{\frac{1}{\sqrt{4\pi}}} \cdot e^{\left(-\frac{(x-5)^2}{4}\right)}}{\cancel{\frac{1}{\sqrt{4\pi}}} \cdot e^{\left(-\frac{x^2}{4}\right)}} = \frac{0.2}{0.8}$$

step2: take ln on both sid and solve for x

$$\ln\left(e^{\left(-\frac{(x-5)^2}{4}\right)}\right) - \ln\left(e^{\left(-\frac{x^2}{4}\right)}\right) = \ln(0.25)$$

$$\rightarrow -\frac{(x-5)^2}{4} + \frac{x^2}{4} = \ln(0.25)$$

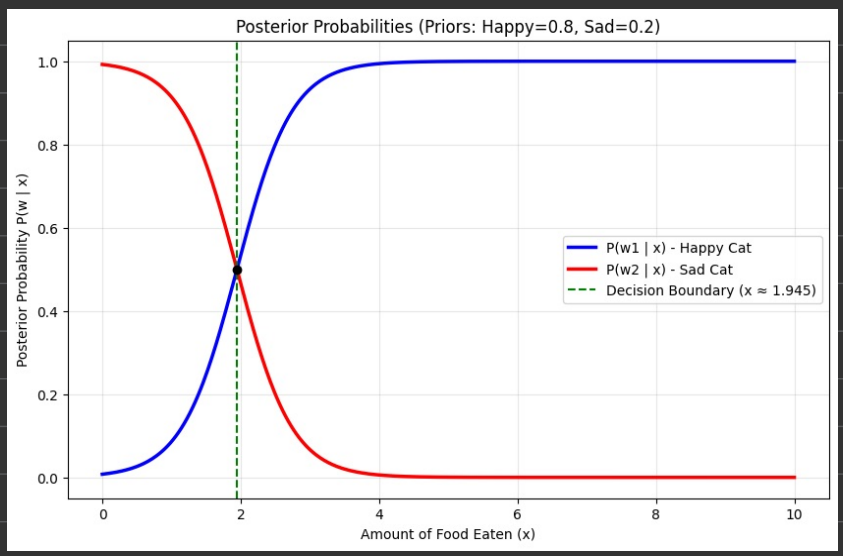
$$\rightarrow x^2 - x^2 + 10x - 25 = 4\ln(0.25)$$

$$10x = 19.454$$

$$x = 1.945$$

∴ The decision boundary is at $x = 1.945$

step3: Define Region



Region 1: $(x < 1.945)$ $P(\omega_2) > P(\omega_1)$
→ Sad Cat

Region 2: $(x > 1.945)$ $P(\omega_1) > P(\omega_2)$
→ Happy cat

Region 3: $(x = 1.945)$ $P(\omega_1) = P(\omega_2)$
→ Decise by yourself

If $x < 2$ it can be in Region 2 but mostly in Region 1

So the answer is Sad Cat but small chance can be Happy cat if $x > 1.945$

Charpter 2

Naive Baye classification

Q6:

- Given:
  - $H(x) = \frac{p(\text{leave})}{p(\text{stay})} \prod_{i=1}^{n} \frac{p(x_i \mid \text{leave})}{p(x_i \mid \text{stay})}$
  - Leave : $H(x) > 1$, stay : $H(x) < 1$
- Find:
  - log-likelihood function of $H(x)$.
  - decision condition.

Solution

<u>step 1</u>: Take ln into $H(x) \rightarrow \ln(H(x)) \sim \mathcal{L}(x)$

$$\mathcal{L}(x) = \ln\left(\frac{p(\text{leave})}{p(\text{stay})}\right) + \ln\left(\prod_{i=1}^{n} \frac{p(x_i \mid \text{leave})}{p(x_i \mid \text{stay})}\right)$$

$$= \underbrace{\left[\ln(p(\text{leave})) - \ln(p(\text{stay}))\right]}_{\text{prior}} + \sum_{i=1}^{n} \left[\underbrace{\ln(p(x_i \mid \text{leave})) - \ln(p(x_i \mid \text{stay}))}_{\text{likelihood}}\right]$$

step 2: state decision condition

we know $H(x) > 1$ is Leave
$H(x) < 1$ is stay

when it log-likelihood

$$\left.\begin{array}{l} \mathcal{L}(x) > \ln(1) \\ \mathcal{L}(x) < \ln(1) \end{array}\right\} \ln(1) = 0$$

$\rightarrow$ $\mathcal{L}(x) > 0$ is Leave
$\mathcal{L}(x) < 0$ is stay

Q7 :

→ Given : • some of $P(x_i \mid \text{attrition})$ can be zero

→ Find : • method to fix this problem

Propose :

Laplace smoothing

→ $$p(x_i \mid c) = \frac{N_{ic} + a}{N_c + a \cdot d}$$

where a is a parameter

d is number of class in feature i

why is this work

→ Recap $p(x_i \mid c) = \frac{N_{ic}}{N_c}$ can be zero

in log-likelihood $\ln(0) \sim -\infty$ can not be use

when come to laplace smoothing a help prevent zero output

a act like an extra of each class an usually 1 this method

make every class have at lease 1 but to be fair with extra

we have to normalize by add the same extra for attrition this

made sure for sum probability of each class equal to 1.